METRIC
MIL-O-60863A(AR)
W/AMENDMENT 5
30 March 2020
SUPERSEDING
MIL-O-60863A(AR)
W/AMENDMENT 4
10 April 1992

MILITARY SPECIFICATION

OXAMIDE

This specification is approved for use by the U.S. Army Combat Capabilities Development Command – Armaments Center (CCDC – AC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of oxamide for use in explosive and propellant formulations as follows:

Type I – Oxamide meeting the particle size requirements given in 3.1.
Type II – Oxamide meeting the granulation requirements given in 3.1.
Type III – Oxamide meeting the requirements given in 3.1.1.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: Commander, U.S. Army CCDC – AC, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or e-mailed to usarmy.pica.ccdc-ac.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-650 - Explosive: Sampling, Inspection and Testing
MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government standards and publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E 11 - Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves (DoD adopted).
ASTM E 300 - Standard Practice for Sampling Industrial Chemicals.
ASTM D 1193 - Standard Specification for Reagent Water

(Copies of these documents are available online at https://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Chemical and physical requirements. Oxamide shall conform to the applicable chemical and physical requirements specified in Table I when tested as specified in the applicable paragraph.

TABLE I. Chemical and physical requirements for oxamide.

Test	Type I	Type II	Test paragraph
Purity, percent, minimum (min.)	99.0	99.0	4.6.1
Moisture content, percent, maximum (max.)	0.1	0.1	4.6.2
Acidity, as oxalic acid, percent, max.	0.05	0.05	4.6.3
Residue, percent, max.	0.05	0.05	4.6.5
Particle size, microns (µm)	4±2	--	4.6.6
Granulation, percent			4.6.7
Thru US Alternative No. 30	--	100.0	
Thru US Alternative No. 325, min.	--	80.0	

3.1.1 Requirements for Type III. The properties of Type III oxamide shall conform to Table I-A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

TABLE I-A. Chemical and physical requirements for Type III oxamide.

Property	Minimum	Maximum	Applicable test paragraph
Assay (NH_2CO_2), %	99.0	--	4.6.1
Moisture content, percent, maximum (max.)	--	0.15	4.6.2
Acidity, as oxalic acid, percent, max.	--	0.045	4.6.3
Particle size, microns (µm)	1	50	4.6.8

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.1.1).

3.3 Workmanship. The manufacturer shall use procedures and controls which assure the oxamide produced does not contain foreign materials such as dirt, rust, paint or metal chips, etc. Compliance with this requirement shall be as specified in 4.6.9.

4. VERIFICATION

4.1 DELETED.

4.2 Classification of inspections. The following inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
c. DELETED.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of three kilograms (6.6 pounds) of the composition obtained by sampling as described in 4.5.1. The samples shall be obtained from a production batch which has been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials shall be from the same sources of supply as will be used in regular production.

4.3.2 Inspections to performed. The sample will be subjected by the Government to any or all of the examinations and tests specified in Table II of this specification and any or all requirements of the applicable drawings.

4.3.3 Rejection. If any sample fails to comply with any of the applicable requirements, the first article shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

TABLE II. First article inspection.

Examination or test	Requirement paragraph	Inspection method paragraph
Purity	3.1	4.6.1
Moisture	3.1	4.6.2
Acidity	3.1	4.6.3
Average particle size	3.1	4.6.5 or 4.6.8
Residue	3.1	4.6.6
Granulation	3.1	4.6.7
Workmanship	3.3	4.6.9

4.4 Quality conformance inspection.

4.4.1 Lot formation. A lot shall consist of a single production batch of not more than 1000 lbs (454 kg) which has been subjected to the same unit chemical or physical mixing process intended to yield a substantially uniform final product. Lot formation shall be in accordance with MIL-STD-1916, Formation and identification of lots or batches.

4.4.2 Examination. Sampling plans and procedures for the classification of defects shall be in accordance with MIL-STD-1916, except that inspection for critical defects shall be 100 percent.

4.5 Testing.

PRECAUTION WARNING

This specification covers sampling, testing and use of toxic or hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Sampling. Using a cement-sampling thief a representative sample of about two pounds in accordance with ASTM E 300 for solids shall be taken from a container randomly chosen from those comprising the lot.

4.5.1.1 Sampling for M864 propellant component. (Type III only). Approximately 500 grams of the composition shall be selected from each batch to be sampled using ASTM Method E 300 for solids. Samples shall be selected for inspection in accordance with the table below. If any sample fails to meet any test requirement, the batch represented by the sample shall be rejected. Each required test shall be performed in duplicate. No composite samples shall be used.

TABLE III. Classification of defects.

Number of containers in batch	Number of containers to be sampled
2 – 200	2
Over 200	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

4.5.2 Acceptance/rejection procedures. The lot represented by the sample shall be tested for compliance with the requirements specified in Table I. The lot shall be accepted if all of the test results comply with the requirements given in Table I. If the test results fail to meet any of the requirements the entire lot shall be rejected.

4.5.3 Classification of defects. The classification of defects shall be as given in Table IV.

TABLE IV. Classification of defects.

Examination or test	Classification
Purity	Major defect
Moisture	Major defect
Acidity	Major defect
Average particle size	Major defect
Residue	Major defect
Granulation	Major defect

4.5.4 Inspection equipment. Commercially available inspection equipment shall be used for all examinations and tests specified in 4.4 and 4.6. The contractor is responsible to have the necessary equipment available, use it correctly and to assure that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes (see 6.4).

4.6 Methods of inspection. The following tests shall be performed using prescribed analytical procedures for replicate determination given in standard analytical textbooks. Unless otherwise specified herein, all chemicals and reagents used in the tests shall be ACS or Reagent Grade. Purified and distilled water in accordance with ASTM Designation D 1193 shall be used as required in all test procedures.

4.6.1 Purity. Accurately weigh to the nearest 0.1 milligram (mg) approximately 0.35 grams (g) of oxamide and transfer to a clean 600 milliliter (mL) beaker. Add 200 mL of 5% sodium hydroxide (NaOH) solution, stir, and place on a steam bath for 1 hour. Cool to room temperature and add about 270 mL of water. Neutralize with 1:1 sulfuric acid and 40 mL 1:1 sulfuric acid in excess. Titrate with standard 0.2 potassium permanganate solution, adding the reagent rapidly at first and then drop by drop as the end point is approached. The end point is a pink color which persists for 30 seconds. Calculate the percent oxamide as follows:

$$\text{Percent oxamide} = \frac{4.403 \text{x} A \text{x} N}{W}$$

Where: A = potassium permanganate solution, mL
N = normality of potassium permanganate solution
W = weight of sample on a moisture-free basis, g

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

4.6.2 Moisture (volatiles). Accurately weigh to the nearest 0.1 mg approximately 5 g of the sample into a tared (with cover) low form 60 mm diameter weighing bottle. Place in a 105±3˚C oven for three hours, cool in a desiccator and weigh. Calculate the percent moisture as follows:

$$\text{Percent moisture} = \frac{100\text{x}(\text{A} - \text{B})}{\text{B}}$$

Where: A = the weight of the sample before drying
B = the weight of the sample after drying

4.6.3 Acidity. Accurately weigh approximately 2 g of the sample to the nearest 0.001 g and quantitatively transfer to a 400 mL beaker. Add about 200 mL of water which has been made neutral to phenolphthalein and a Teflon coated stirring bar and stir on a magnetic stirrer for 5±1 minutes. Filter the mixture through No. 40 Whatman filter paper into a 600 mL beaker. Wash the residue several times with the neutral water and allow the washings to flow into the 600 mL beaker. Add 2 or 3 drops of phenolphthalein indicator to the water and titrate to a light pink end point with 0.01N NaOH which has been standardized to 0.0001N. Calculate the acidity of the oxamide as follows:

$$\text{Percent acidity} = \frac{4.5\text{xAxN}}{\text{W} - (\text{WxM})}$$

Where: A = volume of NaOH solution in mL
B = normality of the NaOH solution
W = weight of the oxamide sample
M = percent moisture of the oxamide expressed as a decimal as determined in 4.6.2

4.6.4 pH. DELETED.

4.6.5 Residue on ignition. Accurately weigh to the nearest 0.1 mg a 2 g sample of the oxamide into a tared platinum crucible. Place the crucible on a supported wire triangle and heat with a Fisher or Meeker or equal gently at first to a red heat. Heat for an additional 15 minutes after all of the organic matter appears to have been completely burned off. Cool the crucible in a desiccator and weigh. Calculate the percent residue on ignition as follows:

$$\text{Residue} = \frac{100\text{ x }(\text{A} - \text{B})}{\text{W} - (\text{WxM})}$$

Where: A = the weight of the crucible and contents after ignition
B = the tare weight of the crucible
W = the weight of sample in g
M = the percent moisture as determined in 4.6.2 expressed as a decimal

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

4.6.6 Average particle size (Type I). The average particle size of Type I oxamide shall be determined in accordance with Method 205.1 of MIL-STD-650. The density of oxamide is 1.667 g/cc.

4.6.7 Granulation (Type II). Nest a U.S. Alternative Sieve No. 30 and a No. 325 (in accordance with ASTM E 11) with the No. 325 sieve on the bottom. Nest a bottom pan below the No. 325 sieve and transfer 100 g accurately weighed to the nearest 0.01 g in the No. 30 sieve. Place a cover on the No. 30 sieve and shake on a mechanical sieve shaker set to produce 300±15 gyrations and 150±10 striker taps per minute for 5 minutes ± 15 seconds. Transfer the oxamide on the sieves and in the bottom pan to appropriate tared weighing containers and weigh each fraction accurately to the nearest 0.01 g and calculate the percent oxamide passing through the No. 30 and No. 325 sieve as follows:

$$\text{Percent oxamide passing thru No. 30 sieve} = \frac{100 - 100 * A}{W}$$

$$\text{Percent oxamide passing thru No. 325 sieve} = \frac{100 - 100 * B}{W}$$

Where: A = the weight of oxamide left on the No. 30 sieve in g
B = the weight of oxamide in the bottom pan in g
W = the weight of the oxamide sample in g

4.6.8 Average particle size. (Type III) The average particle size shall be determined in accordance with Method 205.1 of MIL-STD-650. The sample size shall be 1.7 g.

4.6.9 Workmanship. Visually examine the contents of the container sampled and the sample taken for the presence of foreign materials.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point' packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

6.1 Intended use. Oxamide is intended for use in explosive formulations, and as a component in the M864 and XM1128 propellants.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Material, type, grade, and class required (see 1.2).
c. Issue of ASSIST to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. Whether a first article sample is required (see 3.2).
e. Packaging requirements (see 5.1).

6.2.1 Data requirements. When this specification is used in a procurement contract which incorporates DD Form 1423, acceptance and description sheets prepared in accordance with MIL-STD-1171, or as directed in the contract should be supplied by the contractor.

6.3 Submission of inspection equipment designs for approval. Submit equipment designs, as required, to the Technical Agency (see 6.4)

6.4 Approval of equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the contracting officer to the Technical Agency (see 6.5). This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.5 Technical agency. Picatinny Arsenal is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: Commander, CCDC – AC, ATTN: FCDD-ACE-QMG, Picatinny Arsenal, NJ 07806-5000.

6.6 Subject term (key word) listing.

Igniter
Base bleed
Propellant
Burn rate modifier

6.7 Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodian:
Army – AR

Preparing activity:
Army – AR
(Project 6810-2020-012)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.